

KASHAN, Iran, Islamic Republic ofWMO: **407850**Lat: **33.9667N**Lon: **51.4801E**Elev: **3222**StdP: **13.06**Time Zone: **3.50 (IRN)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.4	28.3	8.6	9.6	44.2	12.5	11.7	45.2	18.7	52.6	14.0	50.8	0.4	210	0.408

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.7	108.0	66.3	105.8	66.0	103.7	65.5	70.8	101.0	69.1	99.5	67.6	98.1	6.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
59.9	86.8	85.7	57.2	78.7	85.1	55.2	73.2	83.0	36.8	100.9	35.2	99.6	33.9	98.2	81.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.3	14.4	11.8	20.4	112.5	6.0	2.5	16.0	114.2	12.5	115.7	9.2	117.1	4.8	118.8		
(5)			19.1	73.8	4.9	4.1	15.6	76.7	12.7	79.1	9.9	81.5	6.4	84.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	67.2	41.4	46.8	56.9	66.2	77.2	87.4	92.1	89.3	81.0	69.1	53.5	44.4	
	DBStd	18.52	7.05	6.75	7.38	6.52	5.59	4.56	3.90	3.50	4.69	6.19	6.74	5.18	
	HDD50	641	272	122	22	1	0	0	0	0	0	0	38	186	
	HDD65	2591	732	511	268	61	1	0	0	0	0	32	347	639	
	CDD50	6922	5	32	236	487	843	1121	1304	1220	929	591	143	11	
	CDD65	3400	0	0	17	97	380	671	839	755	479	159	3	0	
	CDH74	49700	0	0	224	1004	4507	10314	13960	11998	6074	1582	37	0	
	CDH80	30662	0	0	60	295	2199	6598	9643	7937	3346	580	4	0	
Wind	WSAvg	2.7	1.5	2.1	3.2	3.4	3.9	4.0	4.4	3.5	2.3	1.8	1.4	1.2	
Precipitation	PrecAvg	5.6	0.9	0.6	1.0	0.9	0.5	0.1	0.0	0.0	0.0	0.2	0.6	0.7	
	PrecMax	8.7	1.6	2.7	2.4	4.8	2.5	0.5	0.2	0.4	0.1	1.1	2.2	2.1	
	PrecMin	2.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	
	PrecStd	1.5	0.6	0.6	0.7	0.9	0.5	0.1	0.0	0.1	0.0	0.3	0.6	0.5	
	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.1	69.8	86.9	91.6	99.5	108.2	111.6	109.4	102.3	93.5	78.1	66.0
MCWB			47.3	49.2	55.5	59.7	63.0	65.5	67.2	66.9	64.1	61.2	54.2	48.0	
2%		DB	59.4	65.3	79.1	86.4	96.5	105.5	109.3	106.5	99.5	89.6	73.4	62.1	
		MCWB	45.1	47.3	53.0	58.1	62.1	64.9	66.4	66.4	63.4	59.2	53.0	46.7	
5%		DB	56.2	62.3	75.1	82.9	93.5	103.3	107.1	104.3	97.0	86.2	69.5	58.2	
		MCWB	43.4	45.9	51.8	56.9	62.0	64.2	66.3	65.8	62.9	58.7	51.6	45.0	
10%		DB	53.1	59.2	71.1	79.8	90.8	100.7	104.5	102.2	94.7	82.6	65.9	54.5	
		MCWB	42.0	44.6	50.1	55.8	61.1	63.7	66.2	65.3	62.5	57.7	50.0	43.5	
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures		0.4%	WB	49.6	51.4	56.9	62.5	68.6	72.7	73.5	72.7	68.4	63.9	57.1	50.0
			MCDB	60.1	62.7	79.8	82.7	92.0	100.0	103.2	101.4	95.9	86.7	72.2	60.6
	2%	WB	46.6	48.9	54.8	60.2	65.1	69.2	70.9	70.1	66.3	61.7	54.8	48.1	
		MCDB	56.6	60.7	74.4	79.9	90.4	96.3	103.0	100.1	94.1	83.3	69.0	58.1	
	5%	WB	44.6	47.3	53.1	58.6	63.4	66.9	69.3	68.4	64.8	60.2	53.1	46.6	
		MCDB	53.8	59.4	69.9	77.6	88.2	95.4	101.2	99.2	92.6	81.0	65.5	54.4	
	10%	WB	42.6	45.7	51.5	57.3	62.0	65.3	67.9	67.0	63.5	58.7	51.4	45.1	
		MCDB	51.2	57.4	68.0	75.1	86.0	95.2	99.5	98.0	90.9	78.8	62.9	52.2	
	Mean Daily Temperature Range	5% DB	MDBR	18.6	20.3	22.0	22.4	23.6	25.2	24.7	26.5	26.4	24.0	19.8	18.3
			MCDBR	23.6	24.4	27.5	26.7	26.3	27.6	27.2	28.4	28.7	28.0	24.2	24.0
MCWBR			12.8	12.1	11.2	9.6	8.8	9.1	9.1	10.0	10.7	11.6	11.2	12.6	
MCDDBR			21.4	21.7	24.0	22.7	24.9	25.1	25.1	26.0	26.3	25.0	20.9	19.6	
5% WB		MCWBR	12.4	11.4	10.6	9.1	9.8	10.2	10.0	10.8	11.2	11.5	10.1	10.8	
Clear-Sky Solar Irradiance	taub		0.329	0.371	0.443	0.541	0.545	0.535	0.570	0.508	0.447	0.440	0.377	0.328	
	taud		2.327	2.184	1.966	1.774	1.781	1.814	1.757	1.892	2.021	2.019	2.190	2.355	
	Ebn at Noon		277	274	262	242	242	244	234	247	257	246	254	269	
	Edh at Noon		32	41	55	70	70	68	71	61	51	48	36	29	
All-Sky Solar Radiation	RadAvg		963	1265	1638	1885	2225	2578	2472	2319	1981	1488	1060	885	
	RadStd		82	76	110	134	121	84	76	60	56	69	85	88	